

Molnar Single-Notch Benchmark:

Results, Identified Limitation, and Decision Required

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Master's Thesis Progress Review

Thesis Objective

The thesis aims to develop and validate an adaptive-remeshing workflow for phase-field fracture. Before implementing adaptive remeshing, an established fracture benchmark must be reproduced and scientifically assessed.

Why This Benchmark Was Selected

- It is an established Mode-I phase-field fracture example.
- It provides published force-displacement results for comparison.
- It has a known horizontal crack-propagation path.
- It uses the staggered phase-field/displacement implementation being studied.

Plain-Language Current Status

Technical reproduction	Successfully completed
Abaqus simulation	Successfully completed
Crack direction	Qualitatively reproduced
Force-displacement agreement	Mixed, especially after peak load
Phase-field irreversibility	Violations observed between some accepted increments
Overall benchmark assessment	Supervisor decision required

Executive Summary

Candidate v2 is a technically reproducible, paper-matched reconstruction of the published Molnar implementation. It completed successfully and produced a connected, approximately horizontal crack path. The peak-force agreement with the digitized published curve is reasonably close, but the full force-displacement curve shows substantial post-peak disagreement. A targeted diagnostic also found accepted-increment decreases in the phase-field variable. The result should therefore be discussed as a paper-matched reconstruction with an identified irreversibility limitation, not as a strictly irreversibility-clean benchmark.

Detailed numerical evidence, run provenance, diagnostic traces and postprocessing records are available in the project repository upon request.

Actual Benchmark and Simulation Results

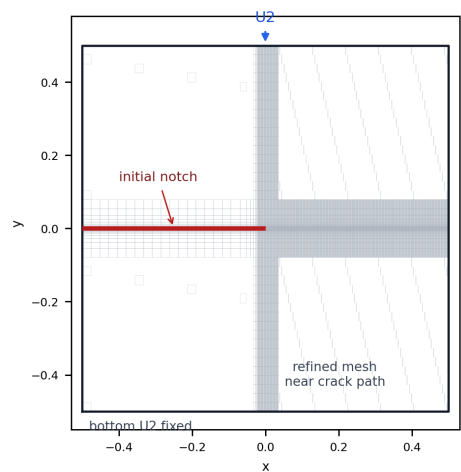


Figure A. Reconstructed Molnar single-notch Mode-I benchmark in normalized/model units, including the applied displacement, boundary conditions, initial notch and locally refined mesh.

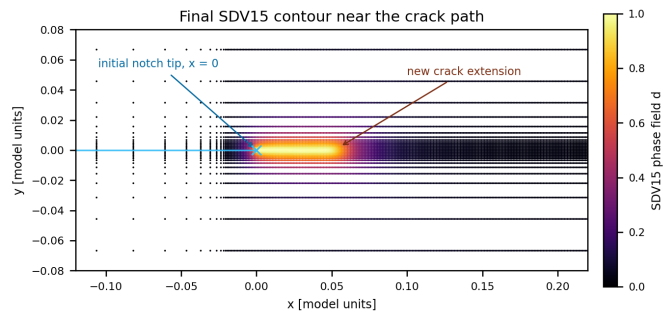


Figure C. Final near-tip SDV15 contour showing a connected, approximately horizontal phase-field extension from the initial notch tip.

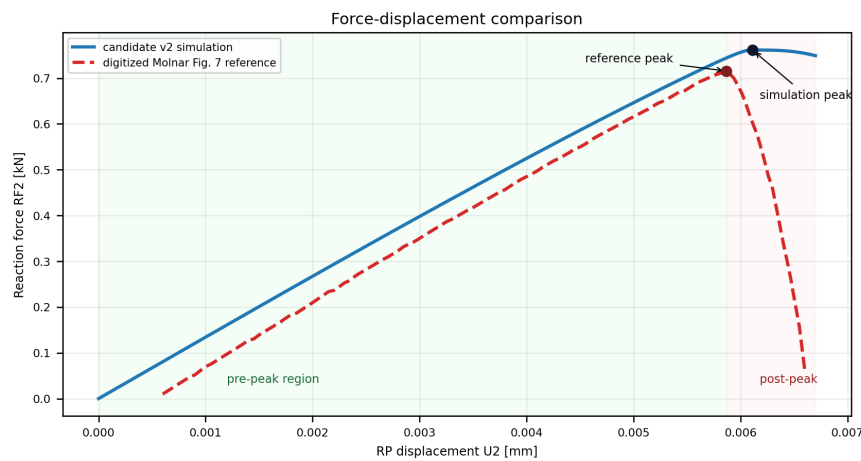


Figure B. Candidate-v2 force-displacement response compared with the approximately digitized Molnar Figure 7 reference. Peak-force agreement is reasonably close, but substantial disagreement remains in the post-peak response.

The published reference was digitized from the paper and is not exact author-provided numerical data.

RF-U Summary

Peak-force error: approximately 6.45%

Peak-displacement error: approximately 4.13%

Full-curve normalized root-mean-square error (NRMSE): approximately 24.57%

Main disagreement: post-peak response

Identified Irreversibility Problem

What SDV15 and SDV16 Mean

- **SDV15**: current phase-field or damage variable, where $d = 0$ is intact and $d = 1$ is fully broken.
 - **SDV16**: maximum stored history-driving energy used to prevent reduction of the fracture-driving history.
- SDV16 remains monotone because it is updated using a maximum-history operation. However, the current implementation does not explicitly enforce $d(n+1) \geq d(n)$, so SDV15 can decrease between accepted increments.

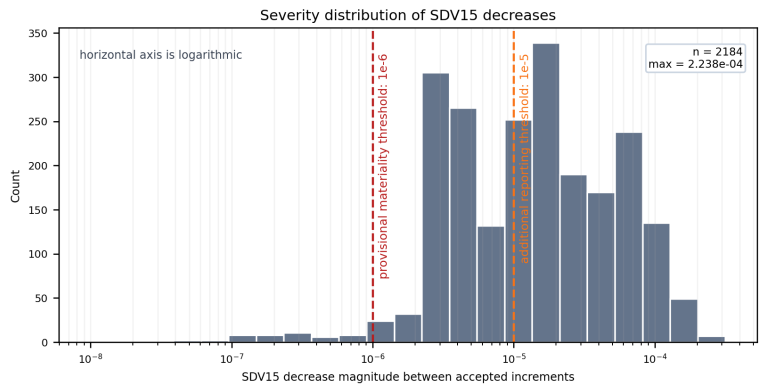


Figure D. Severity distribution of SDV15 decrease magnitudes between accepted increments.

Severity Results

Accepted-increment SDV15 decreases: 2184

Affected element/integration-point locations: 138

Decreases larger than 10^{-6} : 2131

Decreases larger than 10^{-5} : 1362

Maximum decrease: 2.2384×10^{-4}

SDV16 decreases: 0

Timing: all detected decreases occurred after peak load

A diagnostic rerun confirmed that the instrumentation did not change the mechanical response. The normalized RF-U difference from the original simulation was approximately 6.54×10^{-13} .

Interpretation

Candidate v2 is a technically reproducible, paper-matched reconstruction of the published Molnar implementation. However, the phase-field variable decreases between some accepted increments. Because most decreases exceed the provisional project-level materiality threshold of 10^{-6} , which has not yet been approved as a final acceptance criterion, candidate v2 should not be described as satisfying strict pointwise phase-field irreversibility.

The identified issue is specific: candidate v2 remains useful as a paper-matched reconstruction, but it contains a phase-field irreversibility limitation.

Decision Required Before Proceeding

Primary Supervisor Questions

1. Can candidate v2 be retained as the paper-matched reconstructed benchmark baseline, with the phase-field irreversibility issue documented as a limitation?
2. Or should a corrected formulation with explicit irreversibility enforcement be implemented and verified before mesh-convergence and uniform-reference studies begin?

Decision Routes

Option	Meaning	Next action
Accept with documented limitation	Candidate v2 is retained as the paper-matched reconstructed baseline.	Proceed to uniform-reference studies covering mesh size, phase-field length scale and load-increment sensitivity.
Accept temporarily while developing a correction	Candidate v2 remains the paper-matched reconstructed baseline, but a corrected formulation is also developed.	Separate original and corrected formulations clearly
Require correction first	Strict irreversibility is required before further studies	Implement and verify a new formulation using one-element tests before the notch benchmark
Reject candidate v2	The reconstruction is not accepted as the thesis baseline	Reassess the benchmark or select another reference

Recommended Route

My proposed route is to preserve candidate v2 as a technically reproducible, paper-matched reconstruction of the published Molnar implementation, document the identified irreversibility limitation, and proceed with the uniform-reference study. A corrected irreversibility-enforced formulation can be developed as a separate model if strict pointwise irreversibility is required for the later adaptive-remeshing work.

This is a proposal for discussion, not a claim that the route has already been approved.

Reference: Molnar, G., and Gravouil, A. (2017). *2D and 3D Abaqus implementation of a robust staggered phase-field solution for modeling brittle fracture*. Finite Elements in Analysis and Design, 130, 27-38.